

Incident Audit: Deployment Verification Gap

SQL_TEST_01 — Ask-Your-Database Agent (Build 6) — 2026-08-07

Scope: a same-day follow-on to the 2026-08-06 token-burn incident — from the CI fix that closed that incident, through discovering the fix never reached the running browser service, through a narrative-fabrication finding closed the same day, through two further follow-ons surfacing a long-standing history-compounding bug and a related retrieval-routing gap, to a size cap on what the compounding fix still carried forward unbounded, and three more real bugs found reviewing the app's own retry of an earlier question. Closes with an eighth follow-on — the same narrative-fabrication gap Phase 7 closed for `search_docs`, resurfacing on `run_sql_query`, found via live traffic, evidenced with a screenshot, and closed by generalizing Phase 7's own rule rather than patching the symptom again.

Timezone: Pacific (UTC-7), converted from the container's UTC-recorded logs, consistent with the prior audit.

Evidence basis: every timestamp and token count below is pulled directly from `logs/tool_calls.jsonl`, `agent_scratch.chat_rounds`, `docker ps` output, and GitHub Actions run history — not reconstructed from memory.

Executive summary

The eval-harness fix that closed out the 2026-08-06 incident was correct and passed CI cleanly. It never reached the actual running application. The browser-facing Streamlit service had been running continuously since the previous day's last container recreate, and every verification step taken that day — repeated local test runs, a clean CI pass — used *fresh subprocesses* that re-import code from disk on every call. None of them exercised the one thing that mattered: whether the long-running server process itself had picked up the change. It hadn't. Three real user questions were answered by stale code, including one that cost **270,961 tokens** — the single largest token count logged in this project to date, worse than the original incident being fixed. The root cause was a testing-methodology gap, not a code defect: the fix was proven correct in isolation and never proven deployed. Corrected by an actual image rebuild and container recreate, confirmed by a fresh test (9,878 tokens, fully correct), and confirmed a second time independently by the user's own live re-test (15,799 tokens — a 94% reduction from the broken run). That second re-test also surfaced a distinct, unrelated finding: an open gap where the agent can answer an open-ended question from an incomplete semantic-search sample and narrate connective claims the evidence doesn't support — caught correctly by the verifier, so no wrong answer reached the user, but a real gap worth its own fix. Three same-day follow-ons closed the loop further: an automatic pre-commit guard so the documentation search index can't silently go stale again; a fix, surfaced by testing that guard end to end, for a second case where semantic search has no concept of chronological recency and can rank an older document above a genuinely newer one; and a fix for the original narrative-fabrication finding itself, generalizing the same underlying rule that closed the recency gap. Two more same-day follow-ons surfaced afterward, during a routine review of live traffic rather than deliberate testing: a long-standing tiered-memory bug that had been silently compounding stored conversation history roughly 2x every round since the feature's own first use, eventually reaching 938 messages and 176,252 tokens for one question even though that question's own retrieval was entirely correct; and a routing gap where the same kind of question could instead pick a frozen per-

build snapshot over the one continuously-updated file, confidently reporting months-old information as current. Both fixed and independently verified against a freshly rebuilt, freshly recreated container the same day. Two more follow-ons closed out the day. The compounding fix's own full-fidelity window turned out to bound round *count*, not per-round *size* — a risk already named, unmitigated, in this feature's own original planning notes — so one legitimately large tool result still got replayed whole into a follow-up question's context, tripling its cost before that question ran a single tool call of its own; fixed by capping what a round carries forward for others to inherit, not what it sees itself. Finally, retrying an earlier question surfaced three distinct real bugs in one place: a classic unparenthesized AND/OR SQL bug that silently searched the wrong file scope, a main-agent answer silently truncated mid-list by its own token cap with no indication anything was missing, and a verifier that crashed on an oversized payload and showed no badge at all — worse than a wrong one — with its only trace a console line lost to log rotation by the time it was investigated. All three fixed, and the verifier's failure path now writes a durable, structured log entry instead of only printing to a console. An eighth and final follow-on closed the day: reviewing a fresh live exchange found the exact narrative-fabrication gap Phase 7 had already fixed for `search_docs`, resurfacing on `run_sql_query` — a correct 5-item camera list wrapped in fabricated real-world specs, caught precisely by the verifier and evidenced with a live screenshot. Closed the same way Phase 7 closed the original: generalizing the existing grounding rule to name every evidence-gathering tool rather than patching the symptom again, re-verified against a freshly rebuilt container.

Timeline

Phase 0 — Last known-good deploy

2026-08-06, 15:20 PDT

The `askdb_streamlit` container was last rebuilt and recreated the previous day, at the end of the 2026-08-06 token-burn incident (Fix v5, the working default-rule `SYSTEM_PROMPT`). Confirmed directly: `docker ps` showed `CreatedAt: 2026-08-06 15:19:40 -0700 PDT` at the moment this new incident was diagnosed, roughly 20 hours before the fix described below.

Phase 1 — A real CI failure, correctly diagnosed and fixed

11:02 – 11:56 PDT

- A push (closing out the 2026-08-06 incident's documentation) triggered CI at 11:02:21 PDT; the `eval` job failed on a pre-existing, unrelated flaky case ("How many customers are in the database?").
- User: **"Fix the eval issue and any other. We'd like to wrap this project up cleanly."** Investigation found two real bugs, not one: the ambiguous customers question, and a stale hardcoded path in `SYSTEM_PROMPT` (`source_file = 'sql-test-01-cheatsheet.html'`, a bare filename unused since the 2026-08-05 reorg — the real value has been `docs/sql-test-01-cheatsheet.html` ever since). The bare string only ever "worked" locally because 462 orphaned pre-reorg rows were still sitting in the local database, a cleanup flagged and never completed back on 2026-08-05.
- Both fixed, verified stable across multiple fresh local runs (7/7, 5/5, 5/5 verifier cases), committed (`4d8a87f`), pushed, and confirmed via `gh run watch` — CI run `31208524967` passed fully at approximately 11:56 PDT.

Every verification step in this phase was real and passed. That's what makes the next phase worth documenting carefully — a genuinely green CI run and multiple stable local test passes still weren't sufficient evidence that the fix had reached production.

Phase 2 — Stale code answers three real questions

11:29 – 11:40 PDT

Unknown to the person fixing the issue at the time, the user was independently testing the live browser app during this exact window — before the fix's own commit had even finished its CI run. All three questions were answered by the container created the previous day, which had never re-imported any code touched today:

11:29:53	"What information is contained in your docs?"	e66197f0bce0
11:32:29	"Give ma a list of commands used"	8a680229d01f
11:40:27	"Tell me about SQL migrations" — 9 tool calls, 269,042 in / 1,919 out = 270,961 tokens	cba37f905ef6

The largest single-question token count logged in this project. The log shows exactly the bug the Phase 1 fix was meant to close: three separate attempts against the stale `source_file = 'sql-test-01-cheatsheet.html'` (returning nothing), before the model discovered the correct `docs/`-prefixed path on its own — likely from `search_docs`'s own result payload, which includes real `source_file` values — then ran three more narrow, ad hoc `LIKE`-pattern queries against the correct path instead of one direct query. Both problems compounded across all 9 turns via the same stateless-API mechanism documented in the 2026-08-06 audit.

Phase 3 — Diagnosis: the fix was correct, the deployment was not

~12:05 – 12:10 PDT

User: **"Browser app test question with results. Token count still seems extremely high."** First check: `docker exec askdb_streamlit grep source_file agent.py` — the file on disk had the fix. That result was almost misleading on its own; it confirms the file, not the running process. The real check was `docker ps --format "{{.CreatedAt}}"`, which showed the container predated every change made that day.

Root cause: every verification step taken in Phase 1 — `docker exec python -c "..."` test calls, and CI itself — spawns a fresh process that imports `agent.py` from disk on every invocation, so each one correctly picked up the fix. The actual Streamlit server is a single long-running process that imports its dependencies once, at startup. A file changing on disk after that has zero effect on it. Nothing in that day's testing had exercised the one distinction that mattered: *proving code correct* is not the same evidence as *proving a long-running service has reloaded it*.

Phase 4 — Fix and first re-verification

12:10 – 12:13 PDT

- `docker compose build streamlit_app && docker compose up -d --force-recreate streamlit_app` — a real image rebuild and container recreate, not a file patch. Confirmed fresh: `CreatedAt: 2026-08-07 12:10:28 -0700 PDT`, and the fix line present on re-check.
- The 3 chat rounds from Phase 2 removed from `agent_scratch.chat_rounds` (evidence preserved in `logs/tool_calls.jsonl`, which is append-only and untouched) — Build 6's tiered memory would otherwise have replayed that 270,961-token round into the next 3 questions' context regardless of whether the fix worked, the same risk documented in the 2026-08-06 audit.
- Re-ran the exact same question in a fresh process: **9,201 in / 677 out = 9,878 tokens**, one clean `run_sql_query` call, a complete and correct 13-migration list.

Phase 5 — Independent confirmation, and a second, distinct finding

12:13 PDT

12:13:30

"Tell me about SQL migrations" (user's own live re-test, browser) — 1
tool call, 15,408 in / 391 out = **15,799 tokens**

96cd2fb2f2e8

A 94% reduction from the broken run, confirming the deployment fix independently of the test above. But this run took a different path — `search_docs` only, no `run_sql_query` follow-up — and answered from an incomplete top-5 semantic-search sample (4 migrations, not 13), then added connective narrative the evidence never stated: that migrations are "cumulative," "idempotent," and "repeatable," and that `008_add_summary_embedding.sql` "adds vector/embedding support for equipment summaries" (the actual evidence only mentioned an unrelated fix to that file in passing).

The verifier caught this correctly, specifically, and completely — **Unverified**, citing all four unsupported claims individually rather than a generic "doesn't match." Unlike the false positive that opened the 2026-08-06 incident, this is a true positive: a real gap in the answer, correctly identified, that never reached the user unflagged. The system worked as designed here even though the underlying agent behavior — settling for a partial sample and narrating past it on open-ended "tell me about X" questions, as opposed to the more literal "list every X" phrasing `SYSTEM_PROMPT` already steers away from raw search — remains a real, separately-scoped gap, flagged here but intentionally not fixed as part of the incident itself, pending a decision on scope. (Fixed the same day — see Phase 7.)

⚠ Unverified: The proposed answer makes several claims not supported by the tool evidence:

1. **Unsupported claim about migrations being cumulative and idempotent:** The evidence describes what individual migration files do and a specific gap/fix in 012_chat_rounds_schema.sql, but never states that migrations are "cumulative," "idempotent," or "repeatable" in the way described.
2. **Unsupported claim about building fresh databases:** The evidence mentions that "a genuinely fresh database gets the correct grant from one file, not a follow-up migration" in the context of fixing 012_chat_rounds_schema.sql, but this is specific to that one file's fix, not a general principle about how all migrations work.
3. **Unsupported elaboration on 008_add_summary_embedding.sql:** The evidence only mentions this file in passing as having a "CREATE EXTENSION vector fix." The proposed answer claims it "Adds vector/embedding support for equipment summaries," but the evidence doesn't specify what it adds or its purpose—it only indicates there was a fix applied to it.
4. **Overstated completeness claim:** The proposed answer ends by suggesting the cheatsheet provides "a complete, up-to-date list of all migrations used in this project," but the evidence only shows it as containing information about some migrations, not confirmation of completeness.

The evidence supports factual descriptions of specific migration files and a specific bug/fix scenario, but the broader architectural claims about how migrations work in this system go beyond what the evidence actually demonstrates.

The actual verifier output from this question (96cd2fb2f2e8), captured live in the browser — the same reasoning quoted above, unedited.

Before / after

RUN	TOKENS	PATH TAKEN	CORRECTNESS
Stale process (Phase 2)	270,961	3 failed bare-path guesses + 3 narrow re-guesses on the correct path	Eventually correct, at 27x the cost of the fixed version
Fresh subprocess, post-fix (Phase 4)	9,878	One direct, correctly-pathed query	Correct, complete (all 13 migrations)

Live browser, post-fix (Phase 5)

15,799

`search_docs` only, no follow-up

Incomplete + unsupported narrative — correctly flagged
Unverified

Decision log

DECISION	WHY
Rebuild the image and force-recreate the container, not just re-verify the file on disk	The only way to actually change what a long-running process has loaded into memory; a <code>docker cp</code> or <code>grep</code> against the file proves the source is correct, not that the service is running it.
Remove the 3 stale-process chat rounds from persisted history before re-testing	Build 6's tiered memory replays the last 3 rounds in full into every new question's context — leaving them in would have re-cost their token weight regardless of whether the underlying fix worked, the same risk already documented and handled the same way in the 2026-08-06 incident.
Document the <code>search_docs</code> narrative-fabrication finding without fixing it in the same pass	It's a distinct problem from the deployment gap this incident is actually about, correctly caught by the verifier rather than reaching the user silently, and scoping it as its own decision rather than bundling an unrelated fix into an incident report.

Lessons

- A fresh subprocess test and a long-running service are not interchangeable evidence. `docker exec python -c "..."` proves code correctness; only an actual restart of the service being tested proves deployment. Both are needed, and neither substitutes for the other.
- CI passing is real evidence the code is correct in a clean environment — it says nothing about whether a separately-running local or production service has picked up that same code, since CI always builds its own container from scratch.
- The most convincing-looking check ("the file on disk has the fix") can be the wrong one. The right question wasn't "does the source contain the change" but "when was the process that's actually serving requests last started."
- A verifier correctly flagging a bad answer is a successful outcome, not an incident on its own — Phase 5 is worth documenting as a finding, but it's evidence Build 7 is doing its job, not a new failure of the system as a whole.
- `supported: true` only proves the answer is consistent with the evidence gathered for it — it can't judge whether the right evidence was gathered in the first place. Phase 5 and Phase 8b are the same blind spot expressed two different ways: an incomplete sample from the right source, and a complete-looking answer built from the wrong one.
- The same question can take a different path on a different run. Reviewing one successful-looking exchange doesn't prove the prompt reliably produces it every time — Phase 8a's 176,252-token round used the exact right query; the very next fresh attempt at the identical question, once history was clean, used a different and wrong one.

- A cap on round *count* is not a cap on *size* — and this specific gap was already named as a risk, unmitigated, before the feature that caused it had even shipped. Naming a risk in planning notes and actually mitigating it are different steps; only the second one counts.
- A silently-swallowed exception can be a worse outcome than the failure it's protecting against. The verifier crash in Phase 10c left the main answer completely safe, by design — but also left zero durable trace, meaning the only way to actually find it was reproducing the exact API call by hand. A safety net that hides its own failures just relocates the debugging cost to whoever investigates next.
- A model can write SQL that runs without error and returns plausible, on-topic results while still being scoped wrong. An unparenthesized AND/OR mix in Phase 10a didn't fail loudly — it just quietly searched every file instead of one, and the returned content was topically reasonable enough that nothing about the output alone would have raised suspicion.
- A grounding rule fixed for one tool doesn't protect the others by default. Phase 7's fix was correct and held for every `search_docs` question since — but said nothing about `run_sql_query`, so the identical failure mode had an open lane to recur on a completely different tool. The fix that actually closed it, Phase 11, restated the rule as tool-agnostic instead of adding a second tool-specific patch.

Phase 6 — Two follow-on fixes, same day

6a — an automatic guard against the index going stale again

Asked directly: "Can this update process be automatic?" — referring to `docs.chunks` needing a manual `extract_doc_chunks.py` + `embed_chunks.py` re-run after any doc edit, the same class of gap (data this time, not code) as the deployment gap earlier in this incident.

Chosen fix, over two lighter alternatives (a single combined convenience script; a startup staleness warning): a git pre-commit hook, tracked in the repo (`.githooks/pre-commit` , wired in via `git config core.hooksPath .githooks` — not the untracked `.git/hooks/` , which never survives a clone) that re-runs the extract/embed pair whenever a commit touches `docs/sql-test-01-cheatsheet.html` , `docs/project-overview.html` , or any handoff brief. Runs on the host directly (confirmed the host `.venv` already has every required package, and `pg_local` is reachable at `127.0.0.1:5432`) — no container involved. Blocks the commit on failure, the same strictness CI already applies, with `--no-verify` as the standard escape hatch. Tested directly (staged a real pending doc change, ran the hook without committing): correctly detected the change, refreshed 522 chunks, confirmed the 2026-08-07 incident content was queryable immediately afterward.

6b — testing the fix immediately surfaced the finding Phase 5 had left open

Re-running "what was the latest incident reported in the docs" against the now-fresh index still returned **2026-08-06**, not 2026-08-07 — confirmed the new chunk was indexed (a direct query found it), so this wasn't the staleness bug recurring.

`search_docs` ranks by embedding similarity, which has no concept of chronological order; two thematically similar incident callouts don't rank by date just because one is newer. This is the same category of gap Phase 5 flagged and left open (settling for a search result that isn't actually the right kind of answer for the question asked) — "latest/most recent X" needed the identical treatment already given to "list every X."

Fix: `SYSTEM_PROMPT` extended to steer "latest/most recent X" documentation questions toward `run_sql_query` with `ORDER BY chunk_id DESC`, alongside the existing "list every X" guidance — correct specifically because the cheat sheet is append-only (new entries added at the end, never edited in place), so a higher `chunk_id` within that one file reliably means "added more recently." Deployed via a real `docker compose build + up -d --force-recreate`, confirmed fresh via `CreatedAt` before testing — applying Phase 3's own lesson immediately rather than repeating the mistake it describes. Re-tested: 2026-08-07 correctly identified as the latest incident, both incidents correctly ordered, 10,064 tokens. Regression-checked against the full eval suite (7/7, 5/5) before considering it done.

Phase 7 — closing Phase 5's finding: grounding search_docs against incomplete samples

The narrative-fabrication gap flagged in Phase 5 and left open — an open-ended question answered from an incomplete `search_docs` sample, then narrated past what the evidence actually showed — got the same treatment as Phase 6b's fix, generalized further. Root cause common to all three fixes in this incident: a keyword-phrasing trigger ("list every X", then "latest/most recent X") is too narrow, because the real signal isn't the phrasing, it's whether the question is about a category the cheat sheet already tracks as a complete table (every file, every command, every SQL migration) regardless of how it's asked - "tell me about the SQL migrations" needs the same query as "list every SQL migration," even though only one of those literally says "list."

Fix: `SYSTEM_PROMPT` 's SQL-routing rule widened from matching specific phrasings to matching the underlying category - file/command/migration questions route to `run_sql_query` regardless of phrasing. Separately, added an explicit grounding rule for the cases where `search_docs` genuinely is the right tool (narrative/thematic questions with no enumerable category behind them): state only what the retrieved chunks actually say, don't generalize a pattern across the whole system unless a chunk states it explicitly, and say what's missing rather than filling a partial sample with something plausible-sounding.

Re-tested the exact original question ("Tell me about SQL migrations") on a freshly rebuilt and recreated container: this time `run_sql_query` retrieved all 13 migrations in one call (vs. the original 4-migration sample), and the answer's claims all trace directly to retrieved evidence - verified `supported: true`, with the verifier's reasoning specifically confirming each claim against its source chunk rather than a generic pass. 6,959 in / 835 out = 7,794 tokens. Regression-checked against the full eval suite and the verifier eval harness (7/7, 5/5, 5/5) before considering it done.

Phase 8 — Two more same-day follow-ons: a compounding-history bug and a routing gap

8a — the tiered-memory history bug

23:56 PDT

23:56:40

"Latest commits" — 1 tool call, 175,576 in / 676 out = **176,252 tokens**

549cd8bda27e

The single tool call itself was correct — `SELECT chunk_text FROM docs.chunks WHERE source_file = 'docs/sql-test-01-cheatsheet.html' ORDER BY chunk_id DESC LIMIT 100`, exactly the query Phase 6b's fix was meant to produce, and the resulting answer was accurate and complete. The cost came from somewhere else entirely:

`agent_scratch.chat_rounds.full_messages` for this round held **938 messages**. Traced back through every surviving row (as far back as round 86, the earliest still in the table): $6 \rightarrow 10 \rightarrow 20 \rightarrow 40 \rightarrow 76 \rightarrow 144 \rightarrow 274 \rightarrow 506 \rightarrow 938$ — doubling almost exactly every round, present since the tiered-memory feature's own earliest recorded use, not something that started that day.

Root cause: `flatten_history()` keeps the last `MAX_FULL_FIDELITY_ROUNDS` (3) rounds at full fidelity by splicing in each round's stored `full_messages`. But `ask_agent()` never separated a round's *own* new turns from the *inherited* history it started with — every round's stored `full_messages` already contained its own copy of the previous 3 rounds' (already-inherited) histories. Splicing that into the next round nested it again, compounding roughly 2x per round with no bound.

Fix: `ask_agent()` now records where its own turns begin (`own_messages_start`) before appending the user question, and returns/stores only `messages[own_messages_start:]` — each round persists just its own delta, never the inherited prefix. Verified live: 4 sequential real rounds against a freshly rebuilt and recreated container held flat at 4 messages each, with linear (~300 tokens/round) rather than exponential token growth. Two new regression tests added to `test_agent_history.py`, confirmed to fail against the pre-fix code and pass against the fix — checked both ways via a temporary `git stash`, not assumed. Full suite: 29/29 passing. The corrupted `chat_rounds` table (compounding present back to its earliest surviving row) was cleared so the next question would start from a clean baseline.

8b — frozen build snapshots vs. the living cheat sheet

Re-asking the identical question ("Latest commits") immediately after 8a's fix, against a now-clean history — the first time the model had attempted this exact retrieval with no prior turns to anchor against — took a different, wrong path:

`SELECT DISTINCT source_file FROM docs.chunks` , then a direct query against `docs/BUILD_7_FLOW/build7-handoff-brief.html` , on the apparent theory that the highest-numbered build sounds newest. The resulting answer stated **"Build 7 is the latest build,"** silently omitting the 2026-08-05 reorg and every incident and fix since — all of which exist only in

the cheat sheet, never in any handoff brief. The verifier marked this `supported: true` , correctly: the answer was fully consistent with the (wrong) evidence gathered. It has no way to judge whether the right evidence was retrieved in the first place — the same blind spot as Phase 5, expressed a different way: a complete-looking answer built from the wrong source, rather than an incomplete sample from the right one.

Root cause: `SYSTEM_PROMPT` 's SQL-routing rule named specific tracked categories ("every file, every command, every SQL migration") but never named "latest commits/activity/work" as one of them, and never stated the general rule that per-build handoff briefs are frozen the moment that build ships, while only the cheat sheet keeps growing afterward — so a file name that sounds newest is not evidence that it is.

Fix: widened the routing trigger list to include "latest commits/activity/work," and added an explicit rule: the `SELECT DISTINCT source_file` listing exists only to relocate the cheat sheet if its own path ever changes, never to pick a different, better-sounding file instead; handoff briefs are frozen snapshots, the cheat sheet is the only file that keeps growing post-ship. Re-tested the exact failing question on a freshly rebuilt and recreated container: correctly queried the cheat sheet with `ORDER BY chunk_id DESC` , correctly reported the real latest work (2026-08-07 hook and deployment-check entries, 2026-08-06 audit work).
10,393 tokens total. Full suite: 29/29 passing.

Phase 9 — the fidelity window bounds round count, not payload size

A user question — "What's causing the large token usage?" — traced to an already-known, unmitigated risk

Reviewing round 36 ("how many bugs were found while building this app?", 76,403 input tokens) turned up something odd: its own first turn — before it had run a single tool call of its own — already cost **24,143 tokens**. Broken down turn by turn: turn 1 = 24,143, turn 2 = 24,903 (+ round 36's own 2,628-char `search_docs` result), turn 3 = 27,357 (+ round 36's own 9,360-char SQL result). The overwhelming majority of the total was inherited, not earned.

Root cause: the tiered-memory window (Phase 8a's fix) correctly bounds how many *rounds* stay at full fidelity (3) — but nothing bounds the *size* of what a single round carries. Round 35 ("tell me about migrations") had run a legitimate, correctly-scoped query that happened to return 58,824 characters — nearly the entire cheat sheet — and the tiered-memory design replayed that whole result, verbatim, on every turn of round 36, and would have replayed it again for whatever round 37 turned out to be. This exact risk was already named, unmitigated, in this feature's own original Build 6 planning notes: "*The fidelity window bounds round count, not per-round payload size — a much larger table means a single retained `run_sql_query` result could be large regardless of how few rounds are kept raw.*" It took until this incident for that named risk to actually cost something.

Fix: a new `MAX_HISTORY_TOOL_RESULT_CHARS = 3000` cap, applied by `_cap_tool_results_for_history()` at the exact point a round's `full_messages` gets returned/persisted — not at query time. The round that actually gathered the data still reasons over the real, full result live; only the copy carried forward for a *future* round to inherit gets truncated, with a sentinel noting the real size and that re-querying recovers the full data. Verified live by reproducing the exact scenario (a migrations question, then a bugs-found follow-up) against a freshly rebuilt, freshly recreated container: the follow-up's total cost dropped from **76,403 to 23,988 tokens** — a 68.6% reduction for the identical question shape, with round 35's own persisted tool-result content confirmed capped at 3,090 characters instead of 58,824. 3 new tests (pure-function truncation, untouched-when-small, and an end-to-end `ask_agent()` check); full suite 32/32 before moving on.

Phase 10 — three bugs found in one question, on retry

10a — a classic unparenthesized AND/OR bug

Asking the app to retry "how many bugs were found while building this app?" surfaced a model-written query with a real, syntactically-valid-but-wrong bug: `WHERE source_file = '...cheatsheet.html' AND chunk_text ILIKE '%bug%' OR chunk_text ILIKE '%found%' OR chunk_text ILIKE '%incident%'`. SQL evaluates `AND` before `OR`, so the `source_file` filter applied only to the first branch — re-running the exact query directly confirmed all 20 returned rows came from unrelated handoff briefs, **zero** from the cheat sheet the query was supposed to be scoped to. The query ran successfully and returned plausible-looking, topically-relevant content (handoff briefs do document real bugs), which is exactly what made this easy to miss — an error would have been easier to catch than a silently wrong scope.

Fix: an explicit rule added to `SYSTEM_PROMPT` — wrap every `OR` group in its own parentheses whenever a `WHERE` clause mixes `AND` with `OR`, stated with the exact failure mode named, not just "be careful with SQL." Re-tested live: the model now writes `AND (chunk_text ILIKE '%bug%' OR chunk_text ILIKE '%found%' OR chunk_text ILIKE '%surfaced%')`, correctly scoped.

10b — an answer silently truncated mid-list

The same question's final answer opened with "**Total Bugs Found: 14+ confirmed bugs**" and then stopped, mid-list, after only 12 — no ellipsis, no note, nothing indicating anything was missing. Confirmed exactly, not guessed: that turn's `output_tokens` was precisely **1024**, the main agent's own `max_tokens` cap at the time.

Fix: `max_tokens` raised 1024 → 2048 to make this fire less often, and — since a higher cap only narrows the window rather than closing it — a `stop_reason == "max_tokens"` check now appends an explicit *"(This answer was cut off before finishing...)"* note to the answer before it's logged or shown, the same never-silent-truncation rule `MAX_FIELD_CHARS` and `MAX_SQL_RESULT_ROWS` already apply to tool results, extended to the answer itself.

10c — the verifier itself crashed, showing no badge at all

Worse than either bug above: the same question's verifier call never produced a result, and nothing in the UI indicated a verification had even been attempted. Directly reproducing the exact API call (the only way to actually find this — the console print that should have shown it had already scrolled out of the container's log buffer by the time this was investigated) confirmed the cause: the verifier's own `max_tokens=512` cap was hit generating `report_verdict`'s JSON against an unusually large (16,581-character) evidence-plus-answer payload, truncating it after `"supported": false` but before `"reason"` was ever written. `block.input["reason"]` then raised `KeyError`, silently caught by `agent.py`'s broad `try/except` around verification — by design, so a verifier failure can never block the real answer, but with no durable trace of the failure anywhere.

Compounding gap: the safety net worked exactly as designed (the main answer was never blocked or corrupted), but the failure itself left no trace a future debugging session could actually find — only a console `print()`, gone the moment the buffer rotated.

Fix: `verify_answer.py` now uses `.get()` with an explicit fallback reason instead of bracket-indexing a possibly-truncated response, and its own `max_tokens` raised 512 → 1024. A response missing `"supported"` entirely still raises — a worse truncation than the reproduced case, correctly left to the caller's safety net. Separately, a new `log_verification_error()` writes a distinct, structured JSONL entry whenever verification fails for any reason, so a future occurrence is traceable by grepping the log instead of needing to reproduce the exact API call by hand.

Re-tested the exact original question on a freshly rebuilt, freshly recreated container: correctly-parenthesized SQL, a complete answer with no truncation, and the verifier returning a full, real `supported: true` verdict with detailed per-claim reasoning. 8 new regression tests across 3 files (`test_agent_history.py`, the new `test_verify_answer.py`, `test_verification.py`); the key ones confirmed to fail with the exact `KeyError` reproduced above against the pre-fix code, and pass against the fix. Full suite: 39/39.

Phase 11 — closing the loop: outside knowledge added on top of `run_sql_query` results

"What type of cameras do you have?" — the same fabrication gap, on a different tool

18:48 PDT

18:48

"What type of cameras do you have?" — 1 tool call, 10,319 in / 272 out =

07631cb043e0

10,591 tokens

The SQL query itself was correct and narrowly scoped — `SELECT DISTINCT item_name FROM clean.allocation_items WHERE category = 'camera' AND is_accessory = false ORDER BY item_name` — and returned exactly five bare item names, nothing else. The final answer listed those five names correctly, then added a per-item classification next to each one: "Mirrorless camera," "360-degree action camera," "Full-frame mirrorless." None of that appears anywhere in the tool evidence — it's real-world knowledge about these camera models, added on top of the database's own answer. Coincidentally accurate (a Nikon Z6 III genuinely is a full-frame mirrorless camera), but not something the evidence itself supports.

The verifier caught this correctly, specifically, and completely — **Unverified**, naming exactly which claims aren't backed by evidence and confirming the parts that are (the list of five items) separately from the parts that aren't (the classifications). A successful catch, not a new system failure — the same distinction Phase 5 already established.



What type of cameras do you have?



You have 5 types of cameras in your equipment collection:

1. FUJIFILM X-T5 BODY — Mirrorless camera
2. INSTA360 X4 CAMERA — 360-degree action camera
3. NIKON Z6 III MIRRORLESS BODY — Full-frame mirrorless
4. OLYMPUS OM-D E-M1 BODY — Mirrorless camera
5. PANASONIC S5 II BODY — Full-frame mirrorless

These are the base camera bodies (not including accessories like cases, batteries, or lenses). Would you like to know how many of each type you have, or details about their accessories?

⚠ Unverified: The proposed answer adds detailed specifications and categorizations (e.g., "Mirrorless camera," "Full-frame mirrorless," "360-degree action camera") that are not present in the tool evidence. The SQL query output only provides the distinct item names—it does not include any information about camera types, specifications, or classifications. While the list of 5 cameras is accurate based on the evidence, the descriptive details about what type each camera is (mirrorless, full-frame, 360-degree, etc.) represent outside knowledge not supported by the tool results.

> Tools used (1) · 10,591 tokens (10,319 in / 272 out)

Ask about the database or project build



The actual verifier output from this question (07631cb043e0), captured live in the browser.

Root cause: `SYSTEM_PROMPT` 's grounding rule — state only what the evidence actually shows, don't fill gaps with real-world knowledge that merely sounds plausible — was written in Phase 7 specifically for `search_docs` results. It was never generalized to `run_sql_query` results, so nothing told the model that layering outside knowledge on top of a SQL result is the same violation as doing it on top of a search sample.

Fix: the grounding rule widened to name every evidence-gathering tool explicitly, not just `search_docs` — "this grounding rule isn't specific to `search_docs` - it applies to every tool's evidence, including `run_sql_query`," with this exact camera-types answer cited as the confirmed live example directly in the prompt text. Deployed via a real `docker compose build + up -d --force-recreate`, confirmed fresh via `CreatedAt` before testing. Re-

tested the exact original question on the freshly rebuilt container: the model now lists the five item names with no added classifications, and the verifier confirmed `supported: true` — "the proposed answer accurately represents the database findings without overstatement or mischaracterization." Full suite: 39/39 (a prompt-wording change has no unit test of its own to add; validated behaviorally against the live API instead, the only real way to check wording).

Cross-incident comparison: token burn, before and after

Pulling every real logged number across both 2026-08-06 and this incident into one view:

QUESTION	BEFORE	AFTER	REDUCTION	CORRECTNESS
"How many different types of cameras?" (2026-08-06)	177,090	8,024	95.5%	Wrong (4 types) → correct (5)
"How many laptops do you have?" (2026-08-06)	45,469	6,018	86.8%	Wrong (1,501) → correct (482)
"How many types of microphones?" (2026-08-06)	14,801	7,434	49.8%	Correct, false-negative badge → correct + verified
"Tell me about SQL migrations" (2026-08-07)	270,961	7,794	97.1%	Wrong + incomplete → correct, every claim traced to evidence
"Latest commits" (2026-08-07, history-compounding bug)	176,252	10,393	94.1%	Correct but exponentially costly → correct and cheap
"How many bugs were found...?" (2026-08-07, fidelity-window size cap)	76,403	23,988	68.6%	Correct both times — cost only, one inherited round capped from 58,824 to 3,090 chars

"Before" is the worst real logged run for each question, not a hypothetical baseline; "after" is the final, browser- or verifier-confirmed state, not an intermediate fix attempt. The "Latest commits" row skips over one intermediate run, mid-fix: correctly cheap (11,964 tokens) but wrong ("Build 7 is the latest build") — see Phase 8b. It isn't reported as either endpoint because it was neither the worst logged run nor the final confirmed-correct state.

The throughline: every fix that actually held converged on the same shape - resolve the ambiguity once, as data or as a stated default rule, not as a heuristic re-derived from scratch on every question. The fixes that didn't hold on the first attempt were exactly the ones still asking the model

to infer something from context rather than check a fact: the cheap-but-wrong aggregate query (2026-08-06 v1, 11,926 tokens, mislabeled an accessory), and the narrow phrasing-based patches here (2026-08-07, fixed twice before the underlying category-vs-phrasing rule was actually correct). Cost and correctness moved together every time, not against each other - the cheapest wrong answer was never actually cheaper than the eventual correct one, once what it would take to get the model to argue itself out of a bad guess is counted too.